

1. (15 points, 3 points each) Multiple Choice. Only one choice is correct.

(共 15 分, 每小题 3 分) 选择题, 只有一个选项是正确的.

(1) Suppose that  $\alpha_1, \alpha_2, \alpha_3$  are a basis for the nullspace of a matrix  $A$ ,  $N(A)$ . Which of the following lists of vectors is also a basis for  $N(A)$ ?

- (A)  $\alpha_1 + \alpha_2 - \alpha_3, \alpha_1 + \alpha_2 + 5\alpha_3, 4\alpha_1 + \alpha_2 - 2\alpha_3$ .
- (B)  $\alpha_1 + 2\alpha_2 + \alpha_3, 2\alpha_1 + \alpha_2 + 2\alpha_3, \alpha_3 + \alpha_1 + \alpha_2$ .
- (C)  $\alpha_1 + \alpha_2, \alpha_1 + \alpha_2 + \alpha_3$ .
- (D)  $\alpha_1 - \alpha_2, \alpha_2 - \alpha_3, \alpha_3 - \alpha_1$ .

设  $\alpha_1, \alpha_2, \alpha_3$  为矩阵  $A$  的零空间  $N(A)$  的一组基. 下列哪一组向量也是矩阵  $A$  的零空间的一组基?

- (A)  $\alpha_1 + \alpha_2 - \alpha_3, \alpha_1 + \alpha_2 + 5\alpha_3, 4\alpha_1 + \alpha_2 - 2\alpha_3$ .
- (B)  $\alpha_1 + 2\alpha_2 + \alpha_3, 2\alpha_1 + \alpha_2 + 2\alpha_3, \alpha_3 + \alpha_1 + \alpha_2$ .
- (C)  $\alpha_1 + \alpha_2, \alpha_1 + \alpha_2 + \alpha_3$ .
- (D)  $\alpha_1 - \alpha_2, \alpha_2 - \alpha_3, \alpha_3 - \alpha_1$ .

(2) Which of the following statements is correct?

- (A) If the columns of  $A$  are linearly independent, then  $A\mathbf{x} = \mathbf{b}$  has exactly one solution for every  $\mathbf{b}$ .
- (B) Any  $5 \times 7$  matrix has linearly dependent columns.
- (C) If the columns of a matrix  $A$  are linearly dependent, so are the rows.
- (D) The column space and row space of a  $10 \times 12$  matrix may have different dimensions.

以下说法一定是正确的是?

- (A) 如果矩阵  $A$  的列向量线性无关, 那么对任意的  $\mathbf{b}$ ,  $A\mathbf{x} = \mathbf{b}$  有唯一的解.
- (B) 任意  $5 \times 7$  矩阵的列向量一定是线性相关的.
- (C) 如果矩阵  $A$  的列向量线性相关, 该矩阵的行向量也线性相关.
- (D) 一个  $10 \times 12$  矩阵的行空间和列空间可能具有不同的维数.

(3) Let

$$\alpha_1 = \begin{bmatrix} 1 \\ 4 \\ 1 \end{bmatrix}, \alpha_2 = \begin{bmatrix} 2 \\ 1 \\ -5 \end{bmatrix}, \alpha_3 = \begin{bmatrix} 6 \\ 2 \\ -16 \end{bmatrix}, \beta = \begin{bmatrix} 2 \\ t \\ 3 \end{bmatrix}.$$

$\beta$  can be written as a linear combination of  $\alpha_1, \alpha_2, \alpha_3$  if  $t = ( )$

- (A) 1.
- (B) 3.
- (C) 6.
- (D) 9.

设

$$\alpha_1 = \begin{bmatrix} 1 \\ 4 \\ 1 \end{bmatrix}, \alpha_2 = \begin{bmatrix} 2 \\ 1 \\ -5 \end{bmatrix}, \alpha_3 = \begin{bmatrix} 6 \\ 2 \\ -16 \end{bmatrix}, \beta = \begin{bmatrix} 2 \\ t \\ 3 \end{bmatrix}.$$

当  $t = ( )$  时,  $\beta$  可用  $\alpha_1, \alpha_2, \alpha_3$  线性表示.

- (A) 1.
- (B) 3.
- (C) 6.
- (D) 9.

(4) Which of the following statements is correct?

- (A) Suppose that  $EA = B$  and  $E$  is an invertible matrix, then the column space of  $A$  and the column space of  $B$  are the same.
- (B) Let  $A$  be a  $n \times n$  real matrix with rank 1, then  $A^n = cA$ , where  $n$  is a positive integer and  $c$  is a real number.
- (C) Let  $A, B$  be symmetric matrices, then  $AB$  is symmetric.
- (D) If  $A$  is of full row rank, then  $Ax = 0$  has only the zero solution.

以下说法一定是正确的是?

- (A) 设  $E$  为一个可逆矩阵. 如果  $A, B$  矩阵满足  $EA = B$ , 则  $A$  和  $B$  的列空间相同.
- (B) 设  $A$  为秩为 1 的  $n$  阶实矩阵, 则  $A^n = cA$ , 其中  $n$  为正整数,  $c$  为实数.
- (C) 如果  $A, B$  为对称矩阵, 则  $AB$  为对称矩阵.
- (D) 如果矩阵  $A$  为一个行满秩矩阵, 那么  $Ax = 0$  只有零解.

(5) Let  $A$  and  $B$  be two  $n \times n$  matrices. If  $A$  is a non-zero matrix and  $AB = 0$ , then

- (A)  $BA = 0$ .
- (B)  $B = 0$ .
- (C)  $(A + B)(A - B) = A^2 - B^2$ .
- (D)  $\text{rank } B < n$ .

设  $A$  和  $B$  都为  $n$  阶矩阵.  $A$  为非零矩阵, 且  $AB = 0$ , 则

- (A)  $BA = 0$ .
- (B)  $B = 0$ .
- (C)  $(A + B)(A - B) = A^2 - B^2$ .
- (D)  $\text{rank } B < n$ .

2. (25 points, 5 points each) Fill in the blanks.

(共 25 分, 每小题 5 分) 填空题.

(1) Denote the vector space of  $7 \times 7$  real matrices by  $M_{7 \times 7}(\mathbb{R})$ , and let  $W$  be the subspace of  $M_{7 \times 7}(\mathbb{R})$  consisting of skew-symmetric real matrices, then  $\dim W =$ \_\_\_\_\_.

A matrix  $A$  is called skew symmetric if  $A^T = -A$ .

记所有  $7 \times 7$  实矩阵构成的向量空间为  $M_{7 \times 7}(\mathbb{R})$ ,  $W$  为  $M_{7 \times 7}(\mathbb{R})$  中所有斜对称矩阵构成的子空间, 则  $\dim W =$ \_\_\_\_\_.

如果  $A^T = -A$ ,  $A$  就称之为斜对称的.

- (2) Let  $A, B$  be two  $n \times n$  invertible matrices. Suppose the inverse of  $\begin{bmatrix} A & C \\ O & B \end{bmatrix}$  is  $\begin{bmatrix} A^{-1} & D \\ O & B^{-1} \end{bmatrix}$ , where  $O$  is the  $n \times n$  zero matrix. Then  $D =$ \_\_\_\_\_.

设  $A, B$  为两个  $n \times n$  可逆矩阵. 假设  $\begin{bmatrix} A & C \\ O & B \end{bmatrix}$  的逆矩阵为  $\begin{bmatrix} A^{-1} & D \\ O & B^{-1} \end{bmatrix}$ , 其中  $O$  为  $n \times n$  的零矩阵, 则  $D =$ \_\_\_\_\_.

- (3) Let  $A = \begin{bmatrix} a & 1 & 1 & 1 \\ 1 & a & 1 & 1 \\ 1 & 1 & a & 1 \\ 1 & 1 & 1 & a \end{bmatrix}$  with  $\text{rank}(A) < 4$ . Then  $a =$ \_\_\_\_\_.

设  $A = \begin{bmatrix} a & 1 & 1 & 1 \\ 1 & a & 1 & 1 \\ 1 & 1 & a & 1 \\ 1 & 1 & 1 & a \end{bmatrix}$  且  $\text{rank}(A) < 4$ , 则  $a =$ \_\_\_\_\_.

- (4) Consider the system of linear equations:

$$A\mathbf{x} = \mathbf{b} : \begin{cases} x + 2y = 1 \\ x - y = 2 \\ y = -1. \end{cases}$$

The least-squares solution for the system is \_\_\_\_\_.

考虑以下线性方程组:

$$A\mathbf{x} = \mathbf{b} : \begin{cases} x + 2y = 1 \\ x - y = 2 \\ y = -1. \end{cases}$$

该线性方程组的最小二乘解为 \_\_\_\_\_.

- (5) Let  $H$  be the subspace of  $\mathbb{R}^3$  be defined as follows:

$$H = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mid x_1 + 2x_2 + x_3 = 0 \right\}.$$

A unit vector orthogonal to  $H$  is \_\_\_\_\_.

设  $H$  为如下定义的一个  $\mathbb{R}^3$  中的子空间

$$H = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mid x_1 + 2x_2 + x_3 = 0 \right\}.$$

一个和子空间  $H$  正交的单位向量为 \_\_\_\_\_.

3. (24 points) Consider the following  $4 \times 5$  matrix  $A$  with its reduced row echelon form  $R$ :

$$A = \begin{bmatrix} 1 & 2 & * & 1 & * \\ 0 & 1 & * & 1 & * \\ -1 & 1 & * & 3 & * \\ 2 & 0 & * & 1 & * \end{bmatrix}, R = \begin{bmatrix} 1 & 0 & 1 & 0 & 3 \\ 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(a) Find a basis for each of the four fundamental subspaces of  $A$ .

(b) Find the third column of matrix  $A$ .

考虑以下这个  $4 \times 5$  矩阵  $A$  以及它的简化阶梯形矩阵  $R$ :

$$A = \begin{bmatrix} 1 & 2 & * & 1 & * \\ 0 & 1 & * & 1 & * \\ -1 & 1 & * & 3 & * \\ 2 & 0 & * & 1 & * \end{bmatrix}, R = \begin{bmatrix} 1 & 0 & 1 & 0 & 3 \\ 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(a) 分别求矩阵  $A$  的四个基本子空间的一组基向量.

(b) 求出矩阵  $A$  的第三个列向量.

4. (15 points) Let

$$A = \begin{bmatrix} 1 & -1 & -1 \\ 2 & a & 1 \\ -1 & 1 & a \end{bmatrix}, B = \begin{bmatrix} 2 & 2 \\ 1 & a \\ -a-1 & -2 \end{bmatrix}.$$

For which value(s) of  $a$ , the matrix equation  $AX = B$  has no solution, a unique solution, or infinitely many solutions? Where  $X$  is a  $3 \times 2$  matrix. Solve  $AX = B$  if it has at least one solution.

设

$$A = \begin{bmatrix} 1 & -1 & -1 \\ 2 & a & 1 \\ -1 & 1 & a \end{bmatrix}, B = \begin{bmatrix} 2 & 2 \\ 1 & a \\ -a-1 & -2 \end{bmatrix}.$$

当  $a$  为何值时, 矩阵方程  $AX = B$  无解、有唯一解、有无穷多解? 在有解时, 求解此方程. 这里的  $X$  为一个  $3 \times 2$  矩阵.

5. (15 points) Let  $M_{2 \times 2}(\mathbb{R})$  be the vector space of  $2 \times 2$  real matrices. Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, C = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}.$$

Consider the map

$$T : M_{2 \times 2}(\mathbb{R}) \rightarrow \mathbb{R}^3, T(X) = \begin{bmatrix} \text{tr}(A^T X) \\ \text{tr}(B^T X) \\ \text{tr}(C^T X) \end{bmatrix},$$

for any  $2 \times 2$  real matrix  $X$ , where  $\text{tr}(D)$  denotes the trace of a matrix  $D$ .

The trace of an  $n \times n$  matrix  $D$  is defined to be the sum of all the diagonal entries of  $D$ , in other words,

$$\text{tr}(D) = d_{11} + d_{22} + \cdots + d_{nn}.$$

(a) Show that  $T$  is a linear transformation.

(b) Find the matrix representation of  $T$  with respect to the ordered basis

$$v_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, v_3 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, v_4 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

for  $M_{2 \times 2}(\mathbb{R})$  and the standard basis

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

for  $\mathbb{R}^3$ .

(c) Can we find a matrix  $X$  such that  $T(X) = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$ ? If yes, please find one such matrix.

Otherwise, give an explanation.

设  $M_{2 \times 2}(\mathbb{R})$  为所有  $2 \times 2$  实矩阵构成的向量空间. 设

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, C = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}.$$

考虑以下映射

$$T: M_{2 \times 2}(\mathbb{R}) \rightarrow \mathbb{R}^3, T(X) = \begin{bmatrix} \text{tr}(A^T X) \\ \text{tr}(B^T X) \\ \text{tr}(C^T X) \end{bmatrix},$$

对任意的  $2 \times 2$  实矩阵  $X$ , 其中  $\text{tr}(D)$  表示  $n$  阶矩阵  $D$  的迹.

方阵  $D$  的迹是指  $D$  的对角元之和, 也即

$$\text{tr}(D) = d_{11} + d_{22} + \cdots + d_{nn}.$$

(a) 证明  $T$  是一个线性变换.

(b) 求  $T$  在  $M_{2 \times 2}(\mathbb{R})$  的一组基

$$v_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, v_3 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, v_4 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

以及  $\mathbb{R}^3$  的标准基

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

下的矩阵表示.

(c) 是否可以找到一个矩阵  $X$  使得  $T(X) = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$ ? 如果可以, 请求出一个符合要求的矩阵  $X$ . 如果不存在, 请说明理由.

6. (6 points) Let  $A$  be an  $m \times n$  matrix,  $B$  be an  $m \times p$  matrix, and  $C$  be an  $q \times p$  matrix. Show that

$$\text{rank} \begin{bmatrix} A & B \\ O & C \end{bmatrix} \geq \text{rank } A + \text{rank } C,$$

where  $O$  is the  $q \times n$  zero matrix.

设  $A$  为  $m \times n$  矩阵,  $B$  为  $m \times p$  矩阵,  $C$  为  $q \times p$  矩阵. 证明:

$$\text{rank} \begin{bmatrix} A & B \\ O & C \end{bmatrix} \geq \text{rank } A + \text{rank } C,$$

其中  $O$  为  $q \times n$  的零矩阵.